Locality-Stable Extensions and Boundary Laws

LS-2, Collar Typing, and Stabilized Determinacy

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Abstract

This paper isolates the conditional locality layer of the finite relational kernel program. Its purpose is to state LS-2 in a referee-legible way, distinguish raw collar data from realized stabilized collar types, and prove exactly the boundary law needed by later counting arguments. The paper proves no microscopic reconstruction theorem, no continuum limit, and no physical spacetime claim. Its output is narrower and cleaner: under explicit stabilization and locality hypotheses, realized stabilized local outcomes are determined by bounded-radius stabilized collar data.

Contents

1 Introduction

1.1 Why the B-track should stand alone

The flagship kernel paper establishes the first decisive conditional transition of the project: once one assumes LS-2, local stabilized outcome classes become boundary-determined. That step is mathematically modest in syntax but strategically large in consequence. It is exactly the map needed to pass from qualitative locality language to quantitative counting control.

In the flagship paper this material appears in compressed form because the flagship must protect the theorem spine. A separate B-track paper is therefore useful for three reasons. First, it lets the locality package be read without the pressure of packing or bridge claims. Second, it gives the collar-typing conventions enough space to become unambiguous. Third, it makes the negative frontier explicit: LS-2 is a locality law for realized stabilized observables, not a theorem of interior reconstruction, hidden-state recovery, or continuum emergence.

1.2 Inputs from the flagship paper

We take as imported facts the finite kernel, the observational quotient, the existence of refinement chains, and finite stabilization on finite configuration spaces. Those are not reproved here. What this paper adds is the exact conditional structure needed to speak cleanly about collars, local questions, stabilized collar types, and realized outcome dependence.

The logical dependency is strict. No result in this paper bypasses stabilization. No result in this paper converts raw symbolic boundary descriptions into realized outcome laws without an explicit realized-type discipline. This preserves the status partition established in the flagship theorem package.

1.3 What this paper does not claim

This paper does not claim that a boundary collar encodes all microscopic interior structure. It does not claim that equal raw collars imply equal local histories before quotienting. It does not claim a continuum holography statement. It does not identify the collar alphabet with a physical field algebra. It proves only that, under LS-2, the *stabilized observable outcome class* on a region is a function of the *stabilized realized collar type* on a bounded collar.

2 Kernel imports and local questions

2.1 Imported finite setting

We work over a finite configuration space Ω equipped with a refinement program $(\mathcal{T}_n)_{n\geq 0}$ of licensed internal test families. Each stage induces a quotient $\Sigma_n = \Omega/\sim_n$, and finite stabilization yields a stage N such that $\Sigma_n = \Sigma_N$ for all $n \geq N$. We write Σ_∞ for this stabilized quotient.

A point of method matters here. The objects that participate in locality are not raw configurations by default. They are stabilized observable classes, or data extracted from those classes by licensed local procedures. This is the only form in which the paper permits boundary control to be stated.

2.2 Regions and local questions

Let X denote the finite site set of an admissible configuration. A *region* is a finite subset $U \subseteq X$. A *local question on U* is a licensed observable query whose evaluation depends only on admissible data internal to U together with whatever bounded interface data the locality package explicitly allows.

The phrase "depends only on" is not left informal. In this paper it always means dependence at the level of realized stabilized observable classes. Local questions therefore do not inspect unlicensed labels and do not recover information erased by the quotient formalism.

Definition 2.1 (Realized stabilized local outcome)**.** For a fixed region U, a realized stabilized local outcome is an element of a finite set $\mathcal{O}(U)$ of stabilized observable classes obtained by evaluating an admissible local question after stabilization.

Remark 2.2. The word "realized" is important. The paper distinguishes the full symbolic space of possible encodings from the subset that is actually achieved by admissible instances of the theory.

3 Collars and stabilized collar types

3.1 Collars

For an integer radius $r \geq 0$, let $\partial_r U$ denote the admissible radius-r collar around U. The collar is not assumed to be metric in a primitive geometric sense. It is simply the finite interface neighborhood produced by the underlying admissible relational graph structure.

Definition 3.1 (Raw collar description)**.** A raw collar description at radius r is any symbolic encoding of the radius-r interface data around U before quotienting by stabilization or realization.

Definition 3.2 (Stabilized collar type)**.** A stabilized collar type at radius r is the stabilized observable class of the collar configuration $\partial_r U$ under the refinement program.

The second notion, not the first, is the one used in all boundary laws below. Raw collars may distinguish features that are not observationally licensed. Stabilized collar types record only the features surviving the same admissible-test discipline governing the interior.

3.2 Realized collar types

Write $\mathcal{C}_r(U)$ for the set of realized stabilized collar types on $\partial_r U$. This is again a realized set, not a priori the full symbolic assignment space. Later counting arguments require exactly this distinction because injectivity is established into realized collar types, not into arbitrary unrealized strings.

Remark 3.3 (Why realization matters). If one were to count arbitrary symbolic strings before checking realizability, one could create a purely formal codomain much larger than the actually admissible collar sector. Such a move would not invalidate an upper bound, but it would blur the theory's evidentiary discipline. The B-track therefore keeps realized and raw spaces separate from the outset.

4 The LS-2 package

Hypothesis 4.1 (LS-2)**.** There exists a finite radius ρ^* such that for every admissible local question on every finite region U, once refinement has stabilized, the realized stabilized local outcome on U is determined by the realized stabilized collar type on $\partial_{\rho^*} U$.

4.1 Interpretation

LS-2 is a *locality-of-stabilized-observables* statement. It does not assert that interior microstructure is reconstructible from the boundary. It does not assert that the same radius works for nonlocal questions. It does not even assert that the raw collar remembers enough information to recover the stabilized outcome. The claim is narrower: the *stabilized collar type* suffices.

4.2 Why stabilization appears inside LS-2

The dependence on stabilization is not cosmetic. Before stabilization, continuation defects may produce stage-dependent descendants and therefore stage-dependent local answers. In that regime it is not even formally correct to speak of a single local observable class attached to a boundary type. LS-2 is therefore formulated only after the refinement program has settled.

5 Boundary determinacy

Theorem 5.1 (Boundary determinacy under LS-2)**.** *Assume LS-2. Let U be a finite region. If two admissible instances induce the same realized stabilized collar type on $\partial_{\rho^*} U$, then they induce the same realized stabilized local outcome on U.*

Proof. By LS-2, the realized stabilized local outcome on U is determined by the realized stabilized collar type on $\partial_{\rho^*} U$. Therefore equal realized stabilized collar type forces equal realized stabilized local outcome. □

Corollary 5.2 (Boundary law map)**.** *Assume LS-2. For every finite region U there exists a well-defined map*

$$\Phi_U : \mathcal{C}_{\rho^*}(U) \to \mathcal{O}(U)$$

that sends each realized stabilized collar type to the realized stabilized local outcome it determines.

Proof. The theorem shows that any two admissible instances with the same realized stabilized collar type produce the same realized stabilized local outcome. Hence outcome depends only on collar type, which defines Φ_U. □

Proposition 5.3 (Injectivity on realized outcomes)**.** *Assume LS-2. Distinct realized stabilized local outcomes on U require distinct realized stabilized collar types on $\partial_{\rho^*}U$.*

Proof. Suppose two distinct realized stabilized local outcomes arose from the same realized stabilized collar type. The boundary determinacy theorem would then force them to agree, contradiction. □

Remark 5.4 (The true direction of the law)*.* The map Φ_U need not be injective. Several collar types may collapse to the same local outcome. What the theory needs for counting is the converse statement: distinct realized outcomes cannot arise from a single realized collar type.

6 Failure modes and nonclaims

6.1 Without stabilization

If stabilization fails, or if one asks the question before stabilization is reached, then one may face multiple incompatible descendants of the same stage-n class. In that setting a boundary law map may be undefined because the purported outcome has not yet settled into a unique observable class. This is exactly why the flagship paper orders the logic as quotient → stabilization → locality, rather than the reverse.

6.2 Without realized-type discipline

If one forgets the distinction between raw collar strings and realized stabilized collar types, one risks conflating three layers: symbolic encodings, realizable boundary data, and observable boundary classes. The present paper avoids this by treating collar types as stabilized observable classes from the start and then separately recording any symbolic alphabet only in the counting paper.

6.3 No microscopic reconstruction

Nothing proved here implies that the interior of U is uniquely specified by its boundary collar. Many distinct microstates may collapse to the same realized stabilized local outcome. The law proved here is therefore a statement of observable determinacy, not a one-to-one reconstruction theorem.

7 Status ledger

Label	Claim	Status
B1	Precise LS-2 formulation as a realized stabilized boundary law	Conditional
B2	Boundary determinacy from realized stabilized collar type	Conditional
B3	Existence of the boundary law map Φ_U	Conditional
B4	Injectivity on realized outcomes	Conditional
B5	Microscopic reconstruction of raw interiors from collars	Open / not claimed

8 Conclusion

This paper gives the B-track its proper shape. The locality package needed by the dream-paper program is not a vague slogan about the boundary containing the bulk. It is a sharply limited conditional theorem: once stabilization has occurred, realized local observable outcomes are functions of realized stabilized collar types on a bounded collar. That statement is exactly strong enough to support the counting paper and exactly weak enough to remain mathematically honest.

The architectural payoff is substantial. With this paper in place, the flagship kernel remains narrow, the counting layer can proceed without re-explaining locality, and later bridge papers can refer to a clean boundary-law object instead of a compressed paragraph in the flagship.